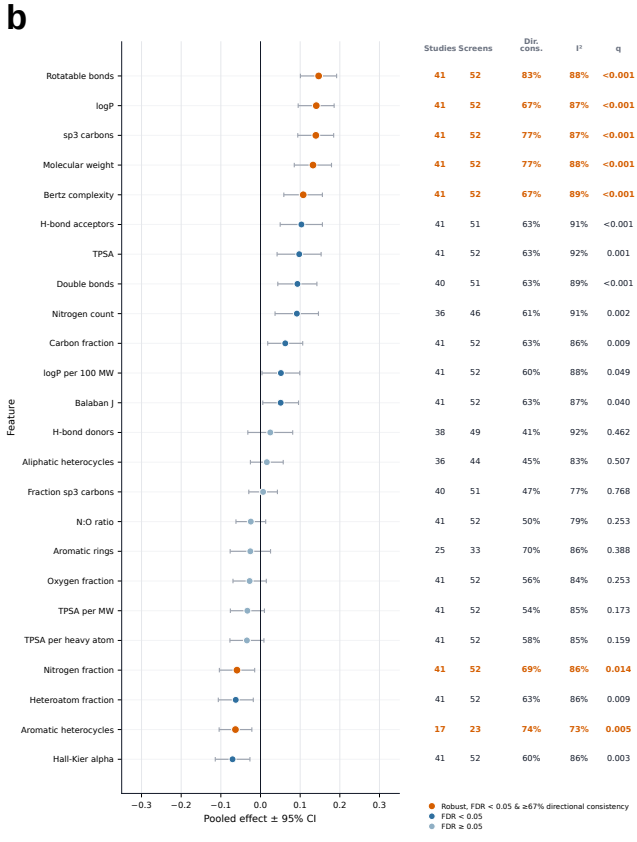
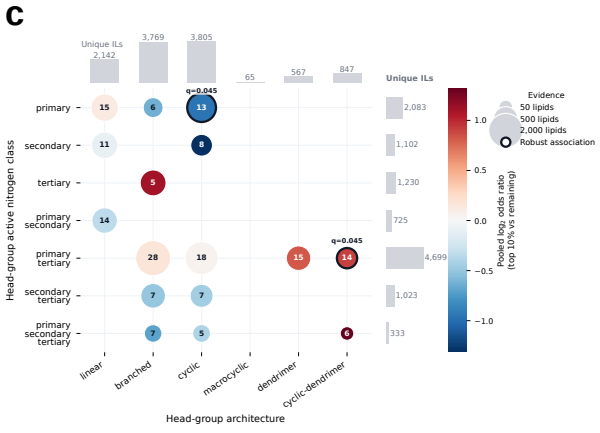
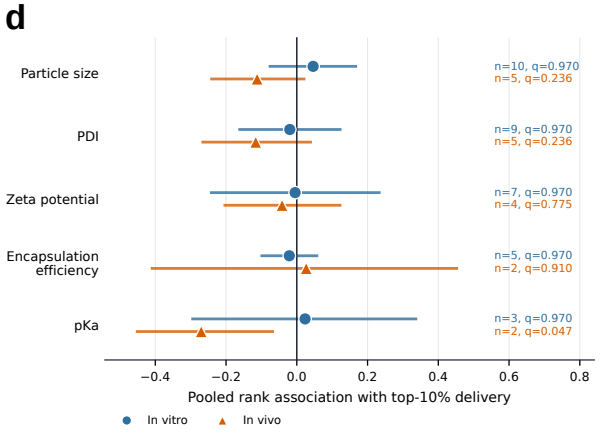
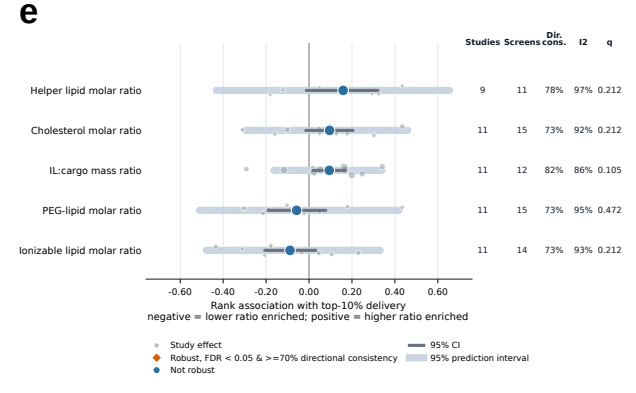
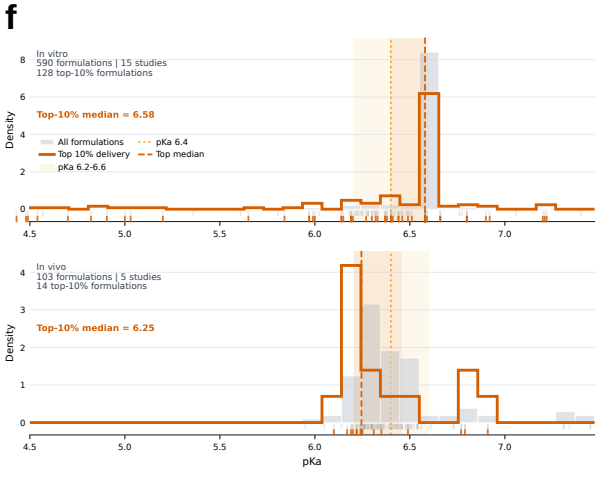

a
Screening contexts
53 screens · 42 studies
study
setting
cargo
model
target
route
metric
>= 30 unique LNPs
>= 30 unique ILs
normalize
Within-screen ranking
0
1
Top 10% highlighted
compare
Cross-study synthesis
Lipid family
Head chemistry
Molecular properties
Formulation
Physicochemical properties
Recurrent design factors

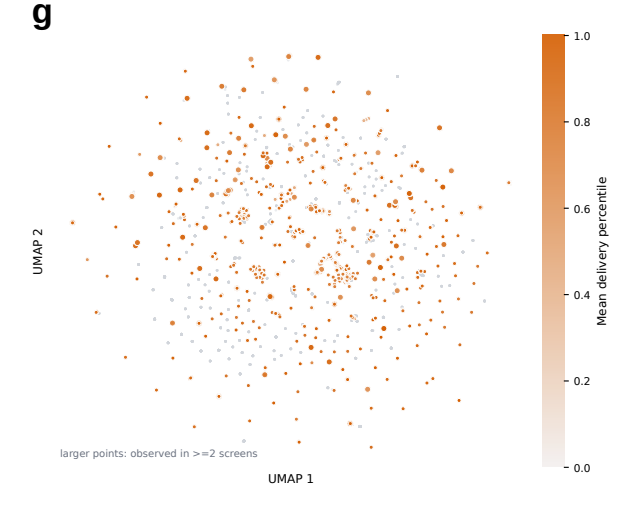

b
Studies
Screens
Dir. cons.
I²
q
Rotatable bonds 41 52 83% 88% <0.001
logP 41 52 67% 87% <0.001
sp3 carbons 41 52 77% 87% <0.001
Molecular weight 41 52 77% 88% <0.001
Bertz complexity 41 52 67% 89% <0.001
H-bond acceptors 41 51 63% 91% <0.001
TPSA 41 52 63% 92% 0.001
Double bonds 40 51 63% 89% <0.001
Nitrogen count 36 46 61% 91% 0.002
Carbon fraction 41 52 63% 86% 0.009
logP per 100 MW 41 52 60% 88% 0.049
Balaban J 41 52 63% 87% 0.040
H-bond donors 38 49 41% 92% 0.462
Aliphatic heterocycles 36 44 45% 83% 0.507
Fraction sp3 carbons 40 51 47% 77% 0.768
N:O ratio 41 52 50% 79% 0.253
Aromatic rings 25 33 70% 86% 0.388
Oxygen fraction 41 52 56% 84% 0.253
TPSA per MW 41 52 54% 85% 0.173
TPSA per heavy atom 41 52 58% 85% 0.159
Nitrogen fraction 41 52 69% 86% 0.014
Heteroatom fraction 41 52 63% 86% 0.009
Aromatic heterocycles 17 23 74% 73% 0.005
Hall-Kier alpha 41 52 60% 86% 0.003
Feature
Pooled effect ± 95% CI
Robust, FDR < 0.05 & ≥67% directional consistency
FDR < 0.05
FDR ≥ 0.05


c
Unique ILs
2,142
3,769
3,805
65
567
847
primary
secondary
tertiary
primary secondary
primary tertiary
secondary tertiary
primary secondary tertiary
q=0.045
Head-group active nitrogen class
linear
branched
cyclic
macrocyclic
dendrimer
cyclic-dendrimer
Head-group architecture
Unique ILs
2,083
1,102
1,230
725
4,699
1,023
333
Evidence
50 lipids
500 lipids
2,000 lipids
Robust association
Pooled log₂ odds ratio (top 10% vs remaining)


d
Particle size
PDI
Zeta potential
Encapsulation efficiency
pKa
n=10, q=0.970
n=5, q=0.236
n=9, q=0.970
n=5, q=0.236
n=7, q=0.970
n=4, q=0.775
n=5, q=0.970
n=2, q=0.910
n=3, q=0.970
n=2, q=0.047
Pooled rank association with top-10% delivery
In vitro
In vivo


e
Studies
Screens
Dir. cons.
I2
q
Helper lipid molar ratio 9 11 78% 97% 0.212
Cholesterol molar ratio 11 15 73% 92% 0.212
IL:cargo mass ratio 11 12 82% 86% 0.105
PEG-lipid molar ratio 11 15 73% 95% 0.472
Ionizable lipid molar ratio 11 14 73% 93% 0.212
Rank association with top-10% delivery
negative = lower ratio enriched; positive = higher ratio enriched
Study effect
Robust, FDR < 0.05 & >=70% directional consistency
Not robust
95% CI
95% prediction interval


f
In vitro
590 formulations | 15 studies
128 top-10% formulations
Top-10% median = 6.58
All formulations
Top 10% delivery
pKa 6.2-6.6
pKa 6.4
Top median
Density
In vivo
103 formulations | 5 studies
14 top-10% formulations
Top-10% median = 6.25
pKa


g
UMAP 2
UMAP 1
larger points: observed in >=2 screens
Mean delivery percentile